

**DEPARTMENT OF**

**COMPUTER SCIENCE & ENGINEERING**

### Experiment 5

**Student Name:** Aditya Kumar Singh

**Branch:** CSE

**Semester:** 6th

**Subject Name:** System Design

**UID:** 23BCS11734

**Section/Group:** KRG 3-A

**Date of Performance:** 26/02/2026

**Subject Code:** 23CSH-314

**1. Aim:** Design a Messenger Application(Similar to WhatsApp / Facebook Messenger): To design and analyze the architecture of a scalable real-time messenger application that supports one-to-one and group communication with high availability and low latency.

### **2. Objective:**

- To implement real-time message delivery using WebSockets.
- To support one-to-one and group messaging.
- To preserve chat history.
- To ensure high availability and reliability.
- To design APIs for user management and chat operations.
- To design the system for large-scale usage (up to 1 Billion users).

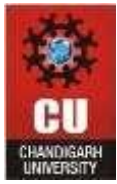
### **3. Tools Used:**

- **Frontend (ReactJS, HTML, CSS, JavaScript, WebSocket)** ○ Built responsive UI with real-time communication support
- **Backend (Node.js / Java / Spring Boot)** ○ Developed scalable REST APIs & WebSocket server
- **Database (MongoDB / MySQL / PostgreSQL)** ○ Managed structured & unstructured data efficiently
- **Redis** ○ Implemented caching & pub-sub for high performance
- **Deployment (AWS / GCP, Load Balancer, Docker)** ○ Cloud deployment with scalability, containerization & traffic distribution

### **4. System Requirements:**

#### **A. Functional Requirements**

- **User Registration**
  - a. POST /user/register
  - b. Fields: userName, email, phoneNumber
- **User Login**
  - a. POST /user/login



## DEPARTMENT OF

## COMPUTER SCIENCE & ENGINEERING

- **One-to-One Messaging**
  - a. WS: /messages/send
  - b. GET /messages/{userId}/{receiverId}
  - c. Pagination support
- **Group Messaging**
  - a. POST /groups/create
  - b. POST /groups/{groupId}/add
  - c. DELETE /groups/{groupId}/remove
  - d. WS: /messages/send
- **Message Types**
  - a. Text messages
  - b. Media messages (images, videos, files)
- **Message History**
  - a. Persistent storage
  - b. Chat list view
- **Read Receipts**
  - a. Delivered
  - b. Seen

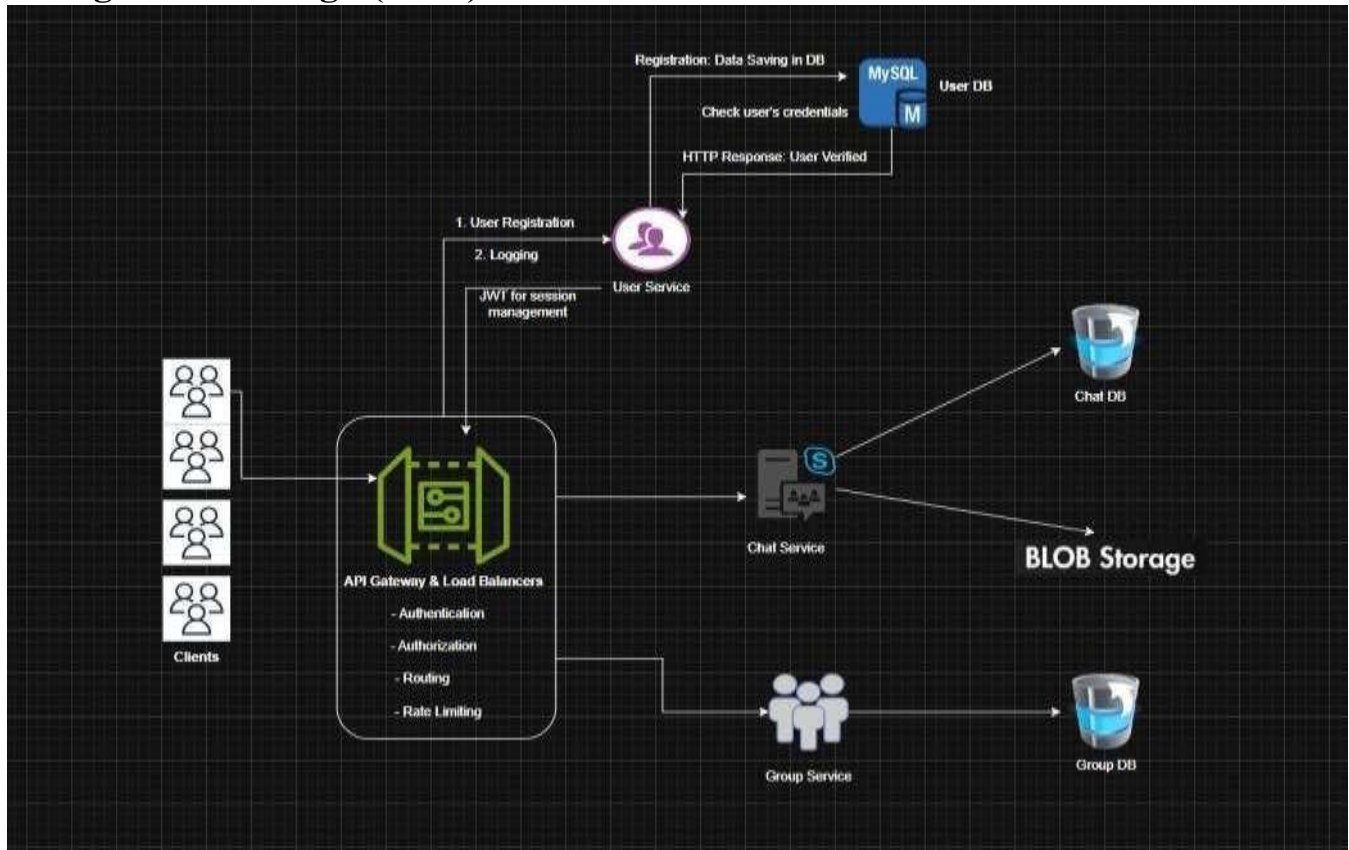
### B. Non-Functional Requirements

- **Scalability**
  - Target: **1 Billion users**
  - 100 messages per user per day
  - $\approx$  100 Billion messages/day
  - Estimated storage:  $\sim$ 100 TB
- **CAP Theorem**

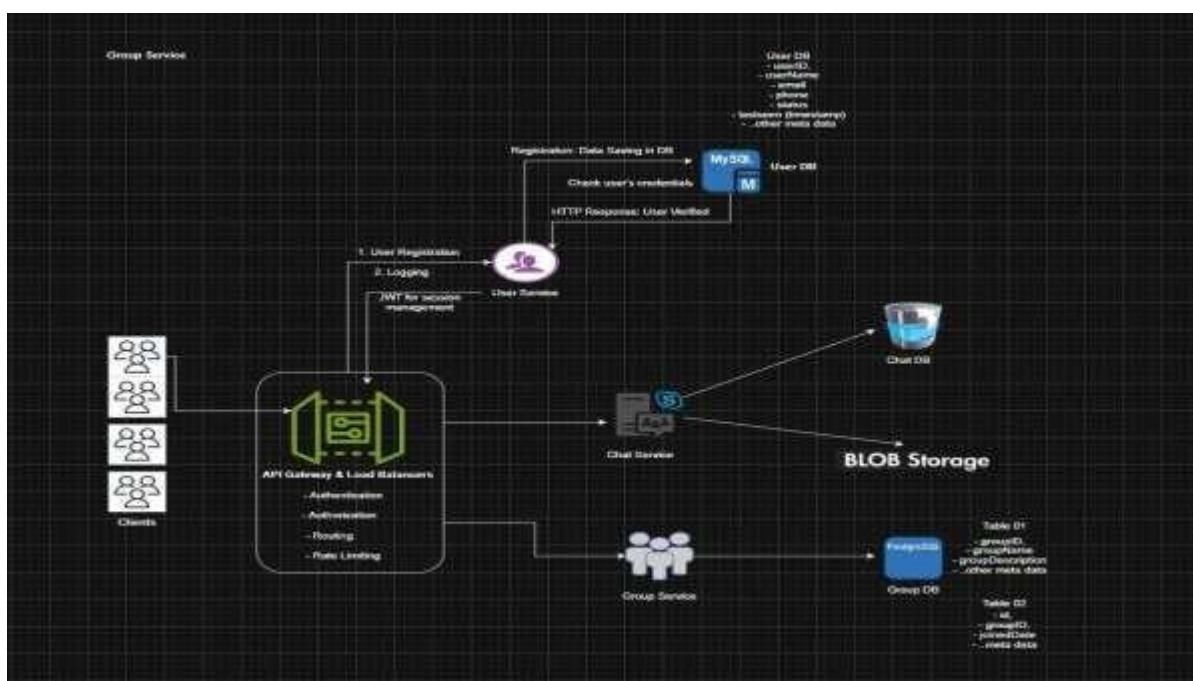
System prioritizes:

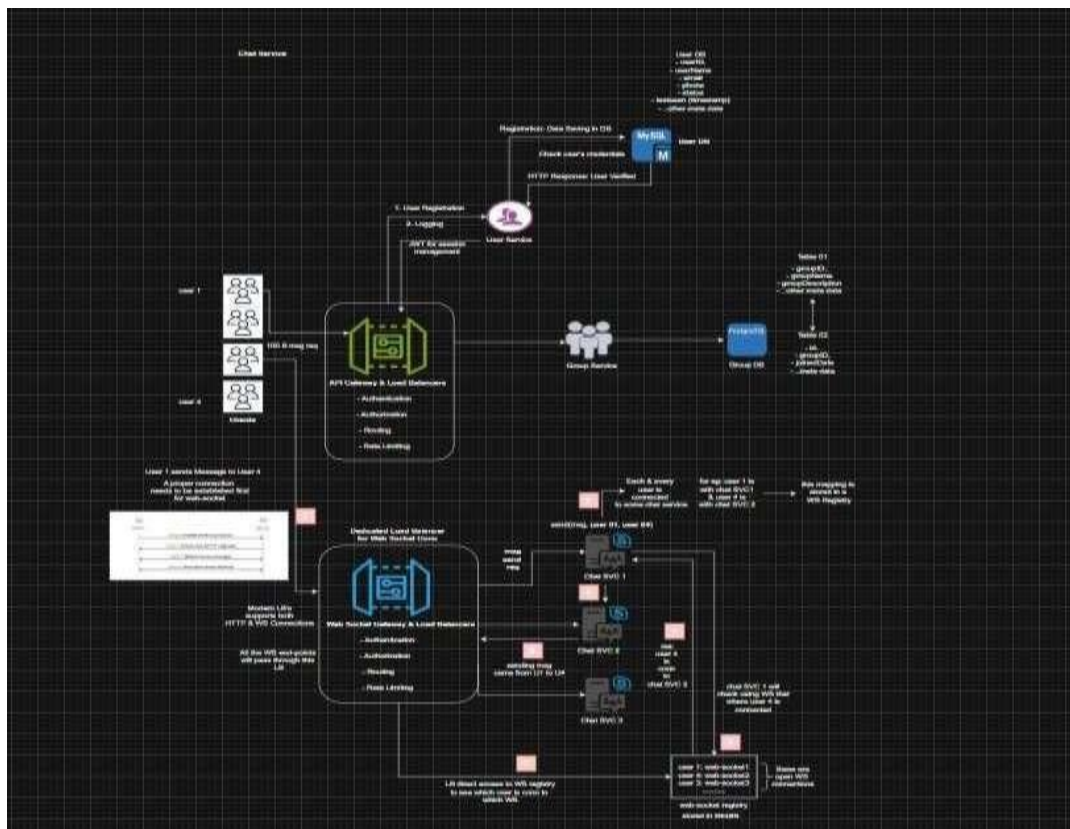
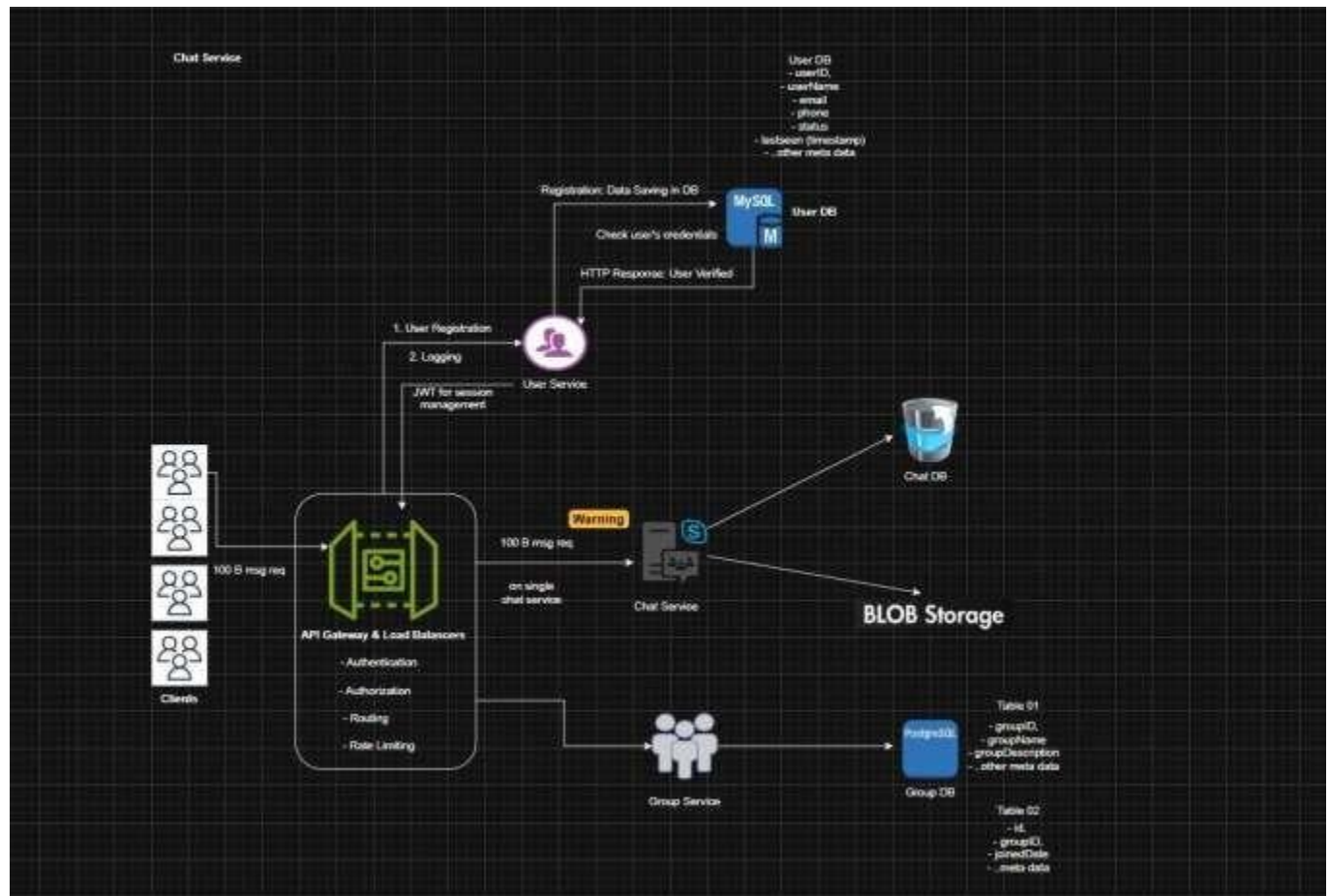
  - High Availability
  - Partition Tolerance
  - Eventual Consistency
- **Latency: 200-300ms**
  - Target message delivery time: **200–300 ms**

## 5. High Level Design (HLD)

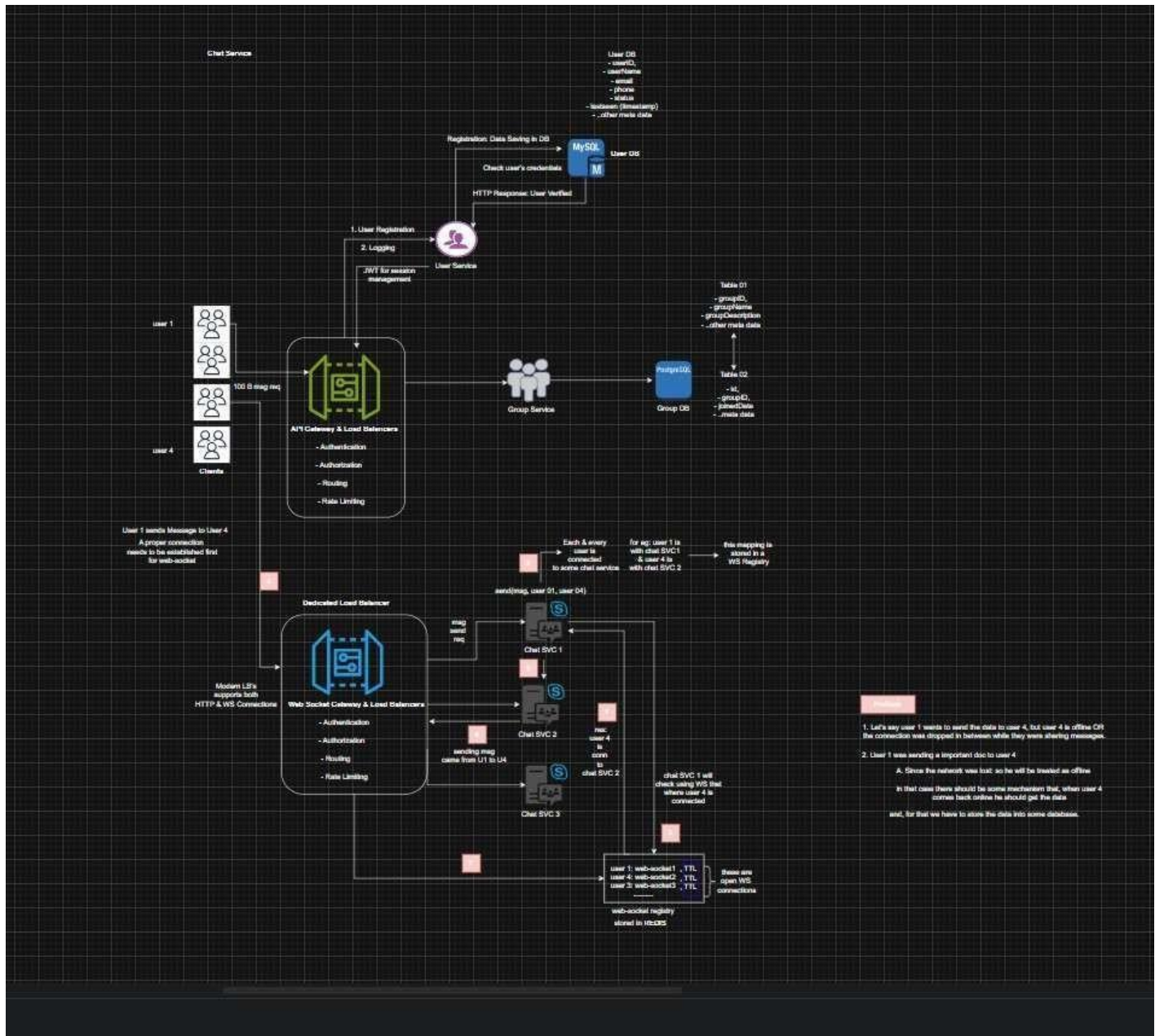


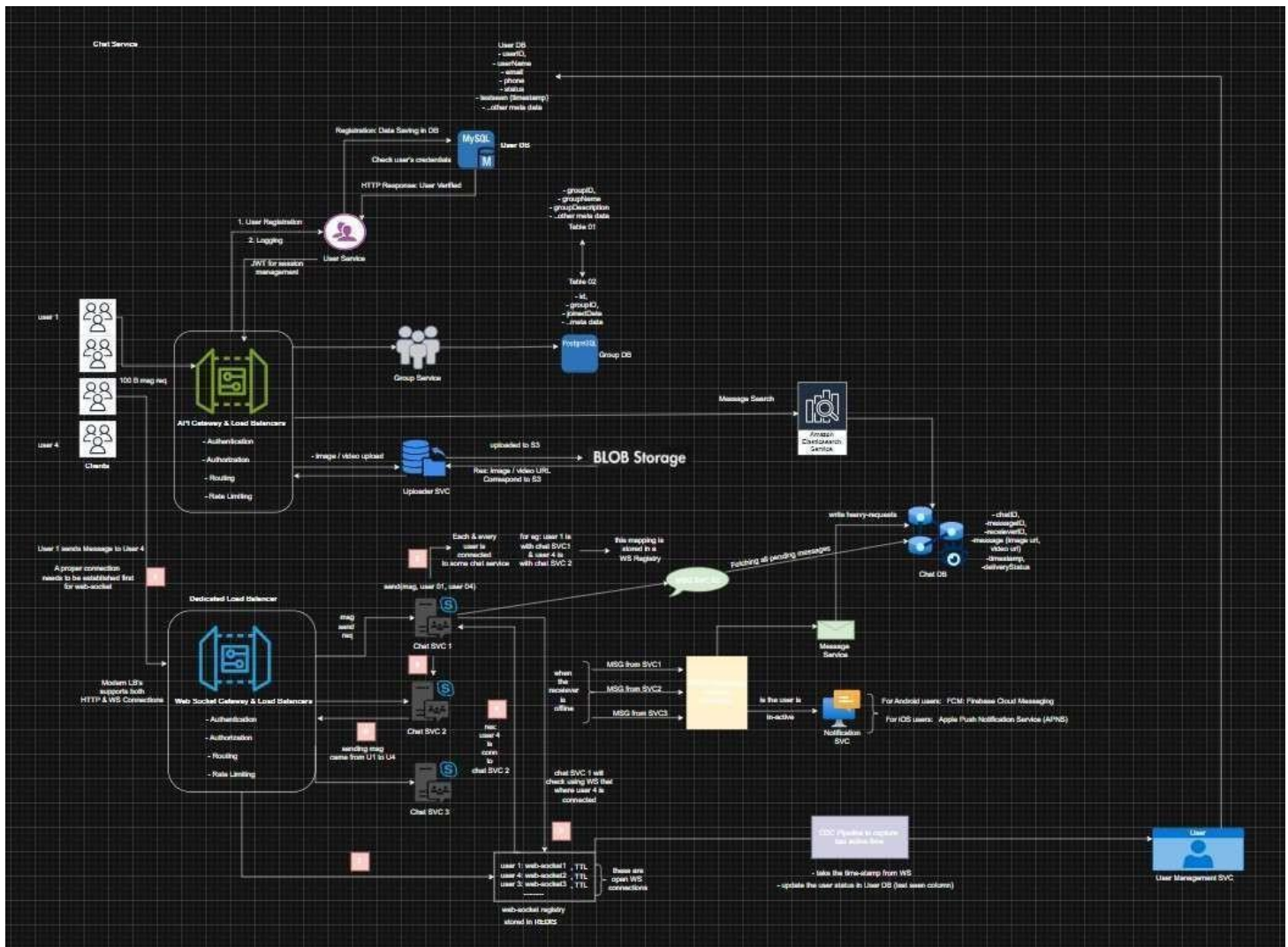
## 6. Low Level Design (LLD):





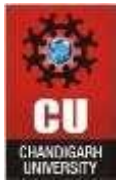






## 7. Scalability Solution

- Horizontal Scaling**  
 Add multiple application servers behind a load balancer to distribute traffic and handle billions of requests efficiently.
- Database Sharding**  
 Partition data based on userID or chatID so that message load is distributed across multiple database servers.
- Caching with Redis**  
 Store frequently accessed chats and recent messages in Redis to reduce database load and improve response time.
- Message Queue System**  
 Use Kafka/RabbitMQ for asynchronous message processing and reliable message delivery.
- CDN for Media Files**  
 Use Content Delivery Network to efficiently deliver images, videos, and files globally with low latency.



**DEPARTMENT OF**

**COMPUTER SCIENCE & ENGINEERING**

### **8. Learning Outcomes (What I Have Learnt)**

- Understand real-time communication using WebSockets.
  - Learn how to design scalable distributed systems.
  - Apply CAP Theorem concepts in real-world system design.
  - Design RESTful APIs and WebSocket endpoints.
  - Analyze trade-offs between consistency, availability, and latency.
- .